

AstroOS

Complete Vedic Astrology Platform with Parashara's Light and Align27 features.

Quick Start

Prerequisites

- Docker & Docker Compose
- Make

Demo Setup (Fastest)

```
make demo
```

This will: 1. Build and start all services (MySQL, Redis, Backend, Frontend) 2. Run database migrations 3. Seed demo data

Access the app at <http://localhost:3000>

Demo credentials: - Email: demo@astroos.com - Password: demo123

Manual Setup

```
# Start services
```

```
make up
```

```
# Run migrations
```

```
make migrate
```

```
# Seed demo data
```

```
make seed
```

Verification

Run complete verification suite:

```
make verify
```

This runs: 1. Database migration up/down test 2. Backend unit tests 3. Frontend build 4. Seed demo data 5. API smoke tests

Available Commands

```
make up           # Start all services
```

```
make down        # Stop all services
```

```
make logs        # View logs
```

```
make migrate     # Run database migrations
```

```
make seed        # Seed demo data
```

```
make demo          # Complete demo setup
make test-backend  # Run backend tests
make test-frontend # Build frontend
make smoke         # Run API smoke tests
make verify        # Run full verification suite
make clean         # Clean up containers and volumes
```

Architecture

Backend (FastAPI + Python)

- Swiss Ephemeris for accurate calculations
- Complete chart calculations (D1-D60)
- 5-level Vimshottari Dasha (MD/AD/PD/SD/Pran)
- Multiple Dasha systems (Yogini, Chara, Ashtottari, Kala Chakra)
- Ashtakavarga (BAV + SAV)
- Yoga detection with rule engine
- Varshaphala (Annual charts)
- Compatibility analysis
- Align27 daily scoring and moments
- RAG with FAISS and OpenAI embeddings
- ML predictions with LightGBM

Frontend (React + TypeScript)

- Modern UI with premium design
- Dashboard for Align27 features
- Chart viewers
- Dasha timelines
- Knowledge base chat
- Multi-profile management

Database (MySQL)

- Complete schema with Alembic migrations
- Proper indexing and constraints
- Audit logging

API Documentation

Once running, access: - API docs: <http://localhost:8001/docs> - Health check: <http://localhost:8001/api/health>

Testing

Backend Tests

```
make test-backend
```

Tests include: - Vimshottari 5-level dasha calculations - Ashtakavarga matrix shapes and invariants - Align27 day scores and moments

Smoke Tests

`make smoke`

Tests: - Health endpoint - Demo setup - Authentication - Profile CRUD

Environment Variables

Create `.env` file in root:

`OPENAI_API_KEY=your_key_here`

Development

Hot Reload

Both backend and frontend have hot reload enabled.

View Logs

`make logs`

Database Access

`docker-compose exec mysql mysql -u astro -p astroos`
Password: astropass

Production Deployment

1. Update environment variables in `docker-compose.yml`
2. Change `SECRET_KEY`
3. Use production OpenAI key
4. Configure proper CORS origins
5. Set up SSL/TLS

Troubleshooting

Services not starting

`make down`
`make clean`
`make up`

Migration errors

```
docker-compose exec backend alembic downgrade base  
docker-compose exec backend alembic upgrade head
```

View backend logs

```
docker-compose logs backend
```

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